⇑Sagar Panchal
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About:

Engineer with hands-on experience developing software for complex machines and real world systems. My career has spanned the full spectrum of software engineering from designing and integrating scalable applications to system enhancements, from developing and managing large-scale codebases to modernize legacy software systems that powers advanced automation platforms. Currently seeking a new opportunity to contribute to building impactful technologies and grow as a problem solver.

Education:

MS, Computer engineering (3.41/4) – Dec-17
New York University NYU
BE, Instrumentation & Controls Engineering (7.04/10) – Jun-14
Gujarat Technological University

Skills:

- PC software based machine automation systems (Java, C#, C, C++)
- Distributed and real time system (currently VxWorks)
- Computer vision applications (experience with Keyence, Cognex and previously OpenCV)
- Git(SourceTree, Bash)
- Maintaining large-scale & legacy codebases
- Custom tools: Python (AI-driven workflows), Web (HTML, CSS, JS), DB integration (MDB, SQL)
- Software for PLC based automation systems (Ladder, FBD), HMI, SCADA

Professional Experience:

Software Engineer III - Automation, Universal Instruments, NY — *Sep 22 - present*

Core Responsibilities:

- Maintain and evolve high-precision, high-speed software modules across the Fuzion SMT platform, ensuring reliability, performance, and production readiness.
- Lead remote production-line diagnostics by analyzing logs, replicating site conditions, performing root-cause investigations and delivering targeted fixes, patches, and release updates.
- Conduct code reviews and enforce software-quality standards across the platform, ensuring robust design, maintainability and consistency in mission-critical automation workflows.
- Enable field and application engineers to successfully deploy and validate applications by providing technical guidance, interpreting machine data, and supplying required configurations or software updates.
- Replicate site issues and lead diagnosis to reach the root cause of problems.

Key contributions:

- Designed, developed, and delivered Selective Placement, Fuzion's first smart pick/buffer/place workflow using customer provided categories and avoiding-priorities, improving product quality and placement intelligence.
- Engineered SBS using Keyence SR-X300, delivering Fuzion's first side-barcode scanning capability, making fuzion scan barcode in all the three X, Y and Z axes, with live-view, traceability, and full error-recovery retuning support.
- Upgraded the AMS workflow from a single point to 13 wafer-level locations, improving calibration robustness and hence maintaining placement accuracy.
- Built the wafer-to-board mapping system that converts customer wafer maps into Fuzion board templates, eliminating hours of manual template creation and enabling direct wafer-data usage.
- Lead Initial implementation of AVP (Advanced Vision Processing), enabling user-defined fiducial finds across ECS, NPI, and Production workflows, increasing alignment flexibility and reducing setup constraints.
- Integrated Cognex QuickBuild (VPP) with MCOS↔QBIT TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Precision-Bonder platform for the first time.
- Delivered multiple platform-level enhancements including the Light Intensity Controller, nested-tray support, isosceles fiducial and configurable park-position, strengthening core system capability.

Automation Engineer , Nor-Cal controls, Ca — *Jan - Jul 22*

- Designed SCADA interfaces and integrated PLC logic with inverter and sensor data to deliver reliable solar power-plant automation including FAT and commissioning support.
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

Software Engineer - Machine Automation, Mini-Circuits, NY — *Feb 18 - Nov 21*

- Evolved machine-automation software across rotary machines with CV Inspection coordinating servo-motion, sensor-driven logic, EVT automation via PCI-e card and PC application based database-supported workflows.
- Enhanced precision laser-measurement stations, semiconductor-packaging machines, and PID-controlled TVAC systems by integrating database-linked classification logic, improving QA reliability and reducing manual effort.
- Helped support modernization of full-stack systems including the public shopping-cart redesign, product-visibility management and lifecycle-tracking tools, increasing platform reliability and removing hours of manual data handling.

Developer Intern, RevMax Fleet Optimization, NY based startup* — *Fall 2017*

- Developed Doppler-radar based pedestrian-counting device and transferred data to back-end services using BLE.
- Deployed sensors at multiple NYC locations and validated field performance

Instructor (ITEST Program), robotics & mechatronics dpt., NYU — *Fall 2017*

- Mentored high-school robotics teams in autonomous-robot workflows for FIRST-Robotics/VEX competitions.
- Taught foundational programming concepts in Arduino & sensor-integration to student robotics teams.

Lead Instructor (CrEST Program), Center for K12 STEM Education, NYU — *Summer 2017*

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
  -Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
  -Wireless Robot Controller - Arduino: https://github.com/sap586/WirelessRobotDay5Activity

Automation Engineer, Prima Automation, Ahmedabad, IN — *Jun 14 - Jan 16*

- Worked with machine-automation systems using Siemens PLC-HMI as control module, provide technical support including software development, testing, FAT/SAT and commisioning support.
- Software development and maintenance (like HMI screens, PLC code, AC-drive configurations, IP-configs, etc) and documentation including control panel layouts made in AutoCAD and electrical drawings made in E-plan.

This document is prepared using LaTeX. Click here to download the source code.